10D 1 MASS SPECTROMETRY OF ORGANIC COMPOUNDS

SPECIFICATION REFERENCE
10.21
PART

LEARNING OBJECTIVES

- Be able to interpret data from mass spectra to suggest possible structures of simple organic compounds.

BACKGROUND TO MASS SPECTROMETRY

You have already come across the principles of mass spectrometry in **Topic 2A.2**. We will now look at how this technique can be used to determine the relative molecular mass of organic compounds. We will also determine the structures of some of these compounds.

The mass spectrum of an element often appears very simple, with a very small number of vertical lines, called peaks. Each peak represents an isotope of the element.

In contrast, the mass spectrum of an organic compound often appears complex, with many peaks. If all the peaks are included, there may be more peaks than there are atoms in a molecule of the compound, so something different is happening.

THE MOLECULAR ION PEAK

The obvious thing to do when first looking at the mass spectrum of an organic compound is to find the peak furthest to the right. This is the one with the greatest m/z value (mass to charge ratio). This peak is the **molecular ion peak** – the result of the organic molecule losing an electron in the mass spectrometer. The equation for this process, using butane as an example, is:

$$C_4H_{10} + e^- \rightarrow C_4H_{10}^+ + 2e^-$$

An electron collides with a butane molecule and knocks out an electron, so forming a positive ion from the molecule. The m/z value of the $C_4H_{10}^+$ peak (58) indicates the relative molecular mass of butane.

OTHER PEAKS

The spectrum may show a very small peak just to the right of the molecular ion peak (sometimes referred to as the $M+1$ peak). This is caused by the presence of a naturally occurring isotope of carbon (^{13}C, rather than the usual ^{12}C isotope) in the molecule. Approximately 99% of all carbon atoms are ^{12}C, with most of the remaining 1% being ^{13}C.

You may have met the ^{14}C isotope, which also occurs naturally in organic compounds. This isotope is radioactive and used in radiocarbon dating. The proportion of ^{14}C atoms in a sample of an organic compound is extremely small and can be ignored in mass spectrometry.

The peaks with smaller m/z values result from fragmentation in the mass spectrometer, rearrangement reactions and the loss of more than one electron. Rearrangement is often unpredictable and will not be considered here, but it does help to explain the large number of peaks in some spectra. The breaking of a carbon–hydrogen bond can occur, but this is not usually described as fragmentation.

FRAGMENTATION IN HYDROCARBONS

Fragmentation is very common and can often be used to work out the structure of an organic molecule.

Consider the breaking of a carbon–carbon bond in the molecular ion formed from a hydrocarbon. Two species are formed. They are:

- another positive ion
- a neutral species (usually a free radical).

EXAMPLES OF FRAGMENTATION

A very simple example of fragmentation is the molecular ion of ethane, which can fragment to form a methyl cation and a methyl radical:

$$(CH_3–CH_3)^+ \rightarrow CH_3^+ + CH_3$$

The ethane molecule is symmetrical, and there is only one carbon–carbon bond in ethane, so you can imagine that the right-hand carbon is just as likely to become the positive ion. The equation for this fragmentation would be:

$$(CH_3–CH_3)^+ \rightarrow CH_3 + CH_3^+$$

You can see that the products are identical, so the spectrum does not depend on how the bond breaks – there will be a peak at $m/z = 15$.

Sometimes the free radical formed is shown with a dot (representing the unpaired electron), and the molecular ion is shown with a dot as well as a positive charge, as in this example:

$$(CH_3–CH_3)^{+\bullet} \rightarrow CH_3\bullet + CH_3^+$$

Now consider propane. There are two carbon–carbon bonds, but they are equivalent – they can both be described as the bond between the central carbon and one of the two terminal carbons. However, there are now two possible fragment ions that can form:

$$(CH_3–CH_2–CH_3)^+ \rightarrow CH_3^+ + CH_2–CH_3$$

$$(CH_3–CH_2–CH_3)^+ \rightarrow CH_3 + (CH_2–CH_3)^+$$

You would therefore expect to see peaks at $m/z = 15$ (the methyl cation) and $m/z = 29$ (the ethyl cation) in its spectrum. These peaks are present, although there are several others that are difficult to explain and are of no help in deducing the structure.

Free radicals are not detected in a mass spectrometer, so all the peaks formed by fragmentation are caused by positive ions.

POSSIBLE INFORMATION

Here is the information you could be given in exam questions.

- A complete mass spectrum – the disadvantage of this is the possible large number of peaks that cannot be used to work out the structure and would be distracting. Another possibility is a complete mass spectrum, but with only the m/z values of the useful peaks marked on the spectrum.
- A simplified mass spectrum showing only the peaks that will help you work out a structure.
- A list of the m/z values of the useful peaks.

With practice, you will be able to work out the structure of an organic compound from this information.

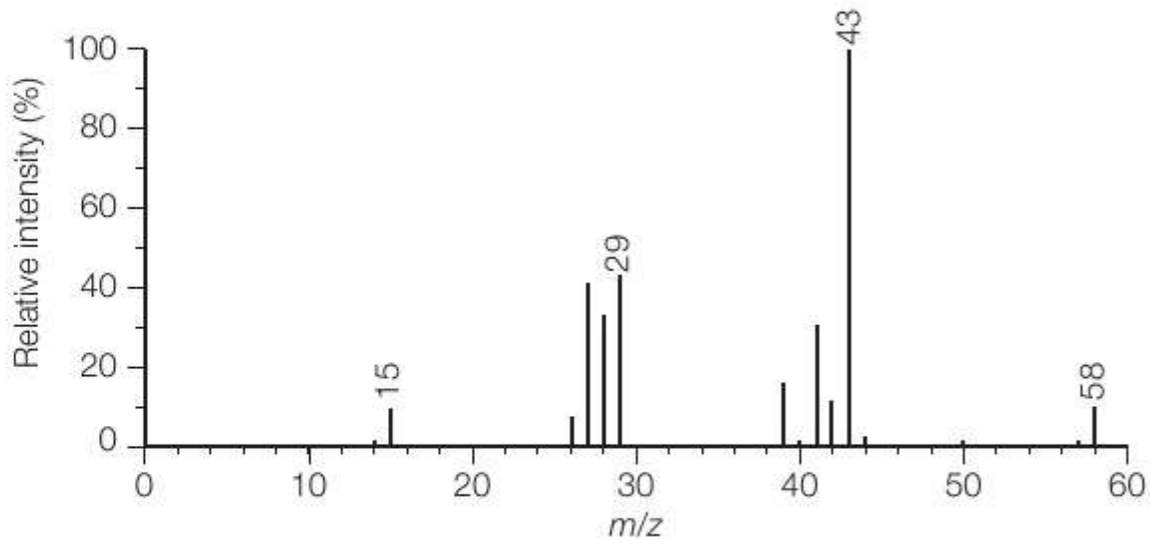
A TYPICAL MASS SPECTRUM

The traditional way to present a mass spectrum is to label the vertical axis as relative intensity (%), always from 0% to 100%. The horizontal axis is labelled m/z (with no units). The horizontal axis usually, but not necessarily, starts from zero and continues to just beyond the molecular ion peak.

The tallest peak is sometimes referred to as the **base peak**.

This base peak represents the ion with the highest abundance, and is shown with a relative intensity of 100%. It represents the most stable fragment.

This is the mass spectrum of butane:



Among the visible peaks are those labelled 15, 29, 43 and 58. **Table A** shows the origin of these peaks.

m/z	ION	NOTES
15	CH_3^+	
29	$(CH_3–CH_2)^+$	
43	$(CH_3–CH_2–CH_2)^+$	This is the most abundant peak.
58	$(CH_3–CH_2–CH_2–CH_3)^+$	This is the molecular ion peak.

table A The origin of the peaks in the mass spectrum of butane.

LEARNING TIP

Be clear about the difference between a peak with a large m/z value and a peak with a large abundance.

The m/z value indicates which ion is present, but the height of the peak indicates its abundance.

CHECKPOINT

1. Explain the origin of the $M+1$ peak seen in some mass spectra.

2. Write an equation to show the formation of the base peak in the fragmentation of the molecular ion of butane.

SUBJECT VOCABULARY

molecular ion peak peak for the species formed from the molecule by the loss of one electron

fragmentation occurs when the molecular ion breaks into smaller pieces

base peak the peak with the greatest abundance

10D 2 DEDUCING STRUCTURES FROM MASS SPECTRA

SPECIFICATION REFERENCE
10.21 PART

LEARNING OBJECTIVES

- Be able to interpret data from mass spectra to suggest possible structures of simple organic compounds.

FRAGMENTATION IN OTHER ORGANIC COMPOUNDS

So far, we have looked only at the mass spectra of alkanes. Now let's look at other organic compounds – those containing oxygen.

In this topic, we will look at two examples of simplified mass spectra and see how they can be used to work out the structures of the compounds responsible for them.

Oxygen is present in many organic compounds, so you need to be familiar with other *m/z* values. You may also come across compounds containing nitrogen (amines) and halogens (halogenoalkanes).

Table A shows some common *m/z* values and possible ions responsible for these peaks.

M/Z VALUE	POSSIBLE IONS
15	CH_3^+
17	OH^+
28	CO^+
29	$CH_3CH_2^+$ and CHO^+
31	CH_2OH^+
43	$CH_3CH_2CH_2^+$ and $CH_3CHCH_3^+$ and CH_3CO^+
45	$COOH^+$ and CH_3CHOH^+
57	$C_4H_9^+$ (this represents four possible structures)

table A Common *m/z* values and the possible ions responsible for these peaks.

LEARNING TIP

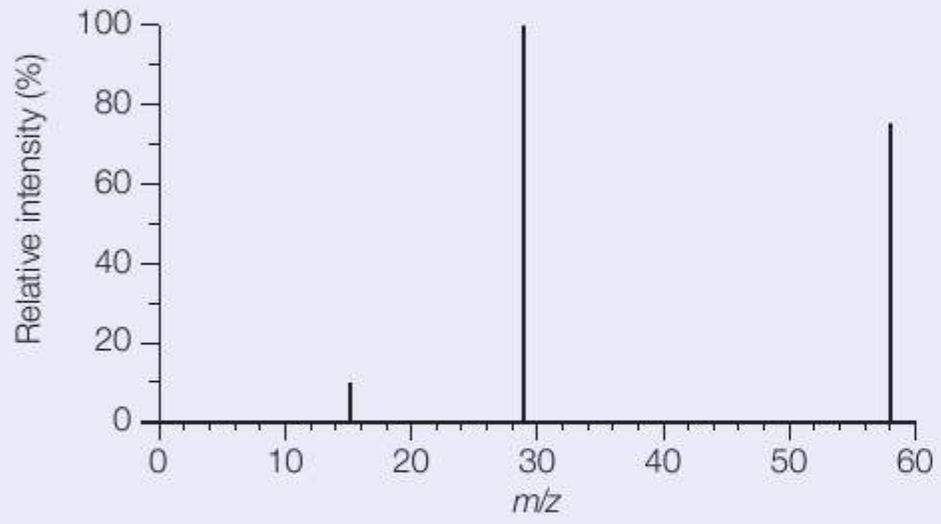
It is sometimes worth comparing the *m/z* values of fragment ions with that of the molecular ion. The difference between the two indicates what has been lost during fragmentation.
For example, if the molecular ion peak is at *m/z* = 60 and there is a fragment ion with *m/z* = 45, the difference is 15, which suggests the loss of a CH_3 group.

WORKED EXAMPLE 1

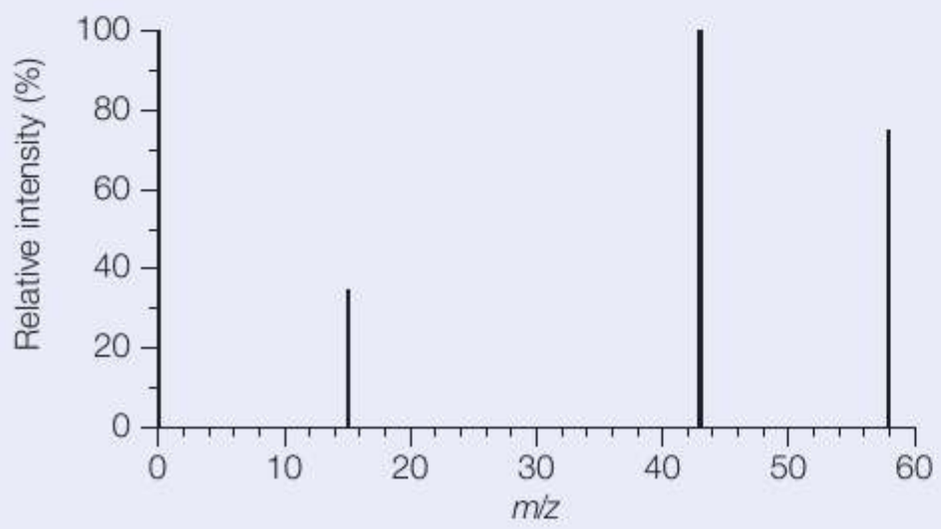
Two compounds, A and B, have the molecular formula C_3H_6O. Their simplified mass spectra are shown below.

Can you deduce the structure of each one?

A



B



INTERPRETATION

Here are the main points that can be deduced from the simplified mass spectra.

- Both mass spectra show a peak at m/z = 58. This corresponds to the molecular ion $C_3H_6O^+$ (relative molecular mass of C_3H_6O is 58.0), but this does not help in deducing the structure.
- Both A and B have a peak at m/z = 15, which is caused by the CH_3^+ ion. As this is present in both, it does not help to distinguish between the two structures.
- One obvious difference is a major peak at m/z = 29 in A, which is not present in B. This could be caused by either $CH_3CH_2^+$ or CHO^+, or by both of them. The structure CH_3CH_2CHO fits perfectly – this is the aldehyde propanal.
- The other obvious difference is a major peak at m/z = 43 in B, which is not present in A. This could not be caused by either $CH_3CH_2CH_2^+$ or $CH_3CHCH_3^+$ because the radical produced at the same time would have a mass of 15 (they must add up to 58), and oxygen has a mass of 16. The other possibility is CH_3CO^+, which when considered with the peak at m/z = 15 (CH_3^+) suggests the structure CH_3COCH_3 – this is the ketone propanone.

You may be asked to write equations to show the formation of the ions you have used in your deduction.

In this example, they are:

A (m/z = 29) $(CH_3CH_2CHO)^+ \rightarrow CH_3CH_2^+ + CHO\cdot$ and $(CH_3CH_2CHO)^+ \rightarrow CH_3CH_2\cdot + CHO^+$

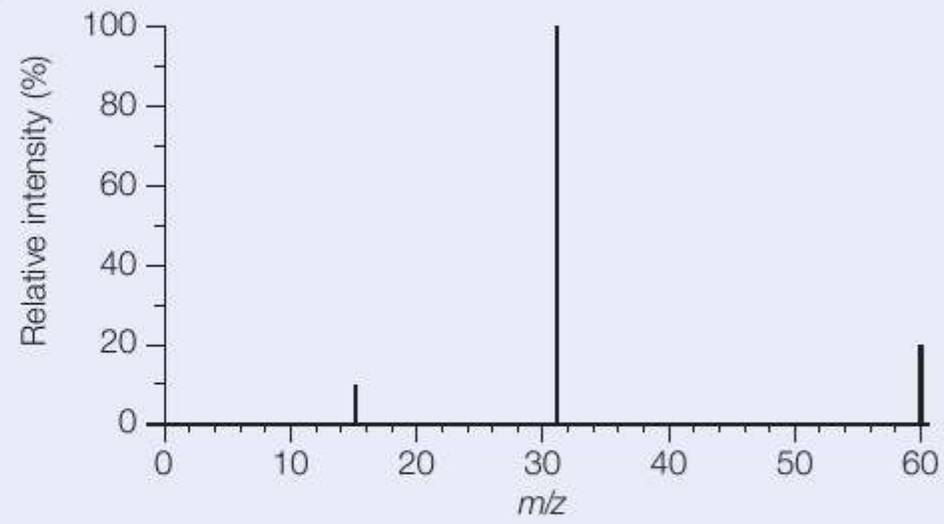
B (m/z = 43) $(CH_3COCH_3)^+ \rightarrow CH_3CO^+ + CH_3\cdot$

WORKED EXAMPLE 2

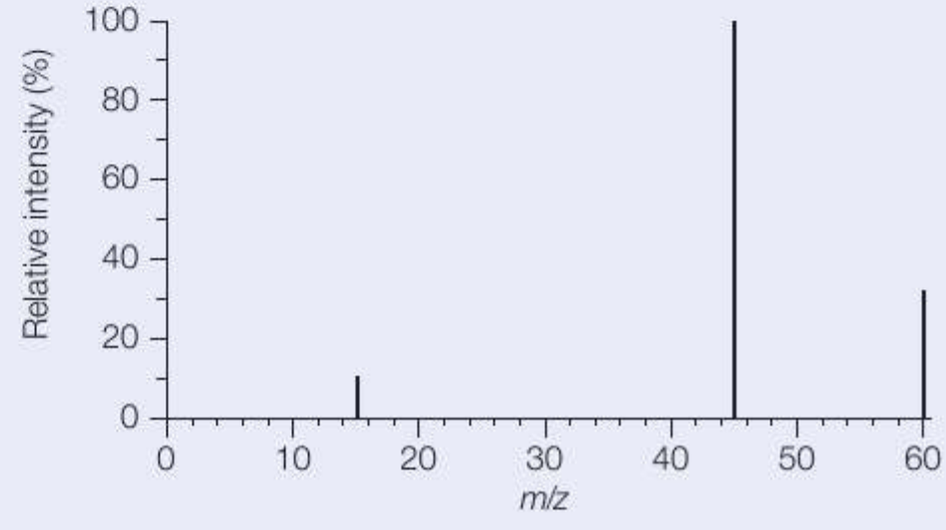
Two compounds, C and D, have the molecular formula C_3H_8O. Their simplified mass spectra are shown below.

Can you deduce the structure of each one?

C



D



INTERPRETATION

Here are the main points that can be deduced from the simplified mass spectra.

- Both mass spectra show a peak at m/z = 60. This corresponds to the molecular ion $C_3H_8O^+$ (relative molecular mass of C_3H_8O is 60.0), but this information does not help in deducing the structure.
- Both C and D have a peak at m/z = 15, which is caused by the CH_3^+ ion. As this is present in both, this information does not help to distinguish between the two structures.
- One obvious difference is a major peak at m/z = 31 in C, which is not present in D. This could be caused by CH_2OH^+. When considered with the peak at m/z = 15 (CH_3^+) and a CH_2 group, this suggests the structure $CH_3CH_2CH_2OH$, which is propan-1-ol.
- The other obvious difference is a major peak at m/z = 45 in D, which is not present in C. This could not be caused by $COOH^+$ because D contains only 1 oxygen atom. The other possibility is CH_3CHOH^+, which when considered with the peak at m/z = 15 (CH_3^+) suggests the structure $CH_3CH(OH)CH_3$, which is propan-2-ol.

CHECKPOINT

1. Write an equation to show the formation of the ethyl cation in the fragmentation of the molecular ion of pentane.

2. Write equations to show the formation of the ions used in the deduction of the structures of C and D (from Worked example 2 above).

10D 3 INFRARED SPECTROSCOPY

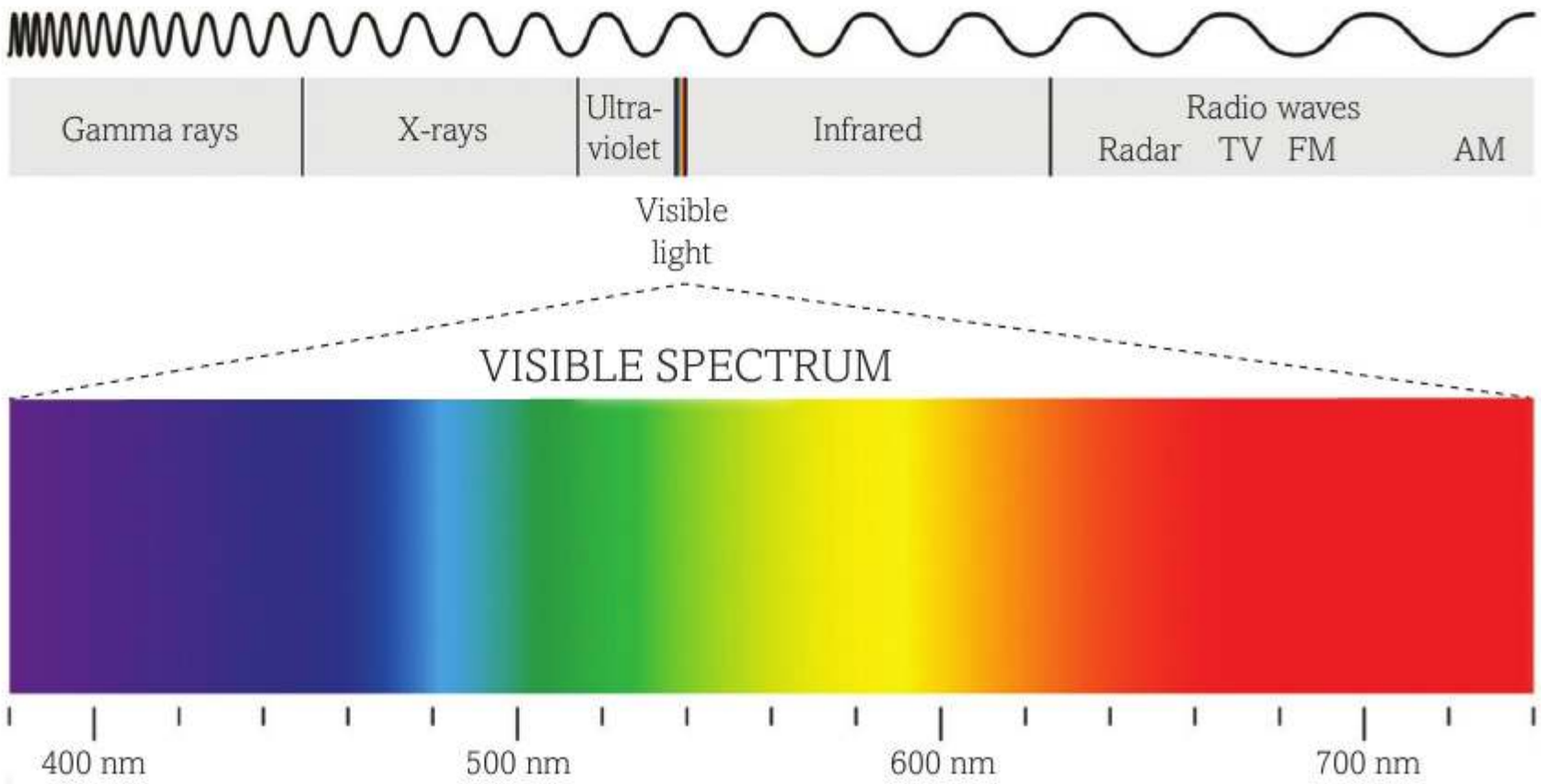
SPECIFICATION REFERENCE
10.22 PART

LEARNING OBJECTIVES

- Be able to use infrared spectra to deduce functional groups present in organic compounds.
- Be able to predict infrared absorptions due to familiar functional groups including wavenumber data.

INFRARED RADIATION

The electromagnetic spectrum of radiation shown in **fig A** includes **infrared radiation**. The 'infra' part of 'infrared' comes from the Latin for 'below', so this radiation has a frequency below, or less than, that of red light.

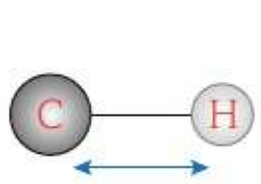


▲ **fig A** You can see how infrared radiation fits into the wider electromagnetic spectrum, next to the red part of the visible spectrum.

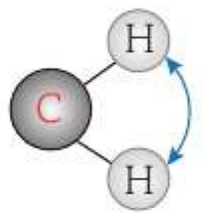
WHAT HAPPENS WHEN MOLECULES ABSORB INFRARED RADIATION

The importance of infrared radiation in chemistry is that it is absorbed by molecules and causes two possible effects, both described as vibrations. These effects are:

- **stretching** – where the bond length increases and decreases
- bending – where the bond angle increases and decreases.



The C–H bond stretches when it absorbs infrared radiation.



The C–H bond bends when it absorbs infrared radiation.

In this book, only stretching vibrations will be considered.

When a molecule absorbs infrared radiation, the amount of energy absorbed depends on:

- the length of the bond
- the strength of the bond
- the mass of each atom involved in the bond.

The absorption of infrared radiation is linked to changes in the polarity of the molecule, so simple non-polar molecules (such as H_2 and Cl_2) do not absorb infrared radiation.

WHAT DOES AN INFRARED SPECTRUM LOOK LIKE?

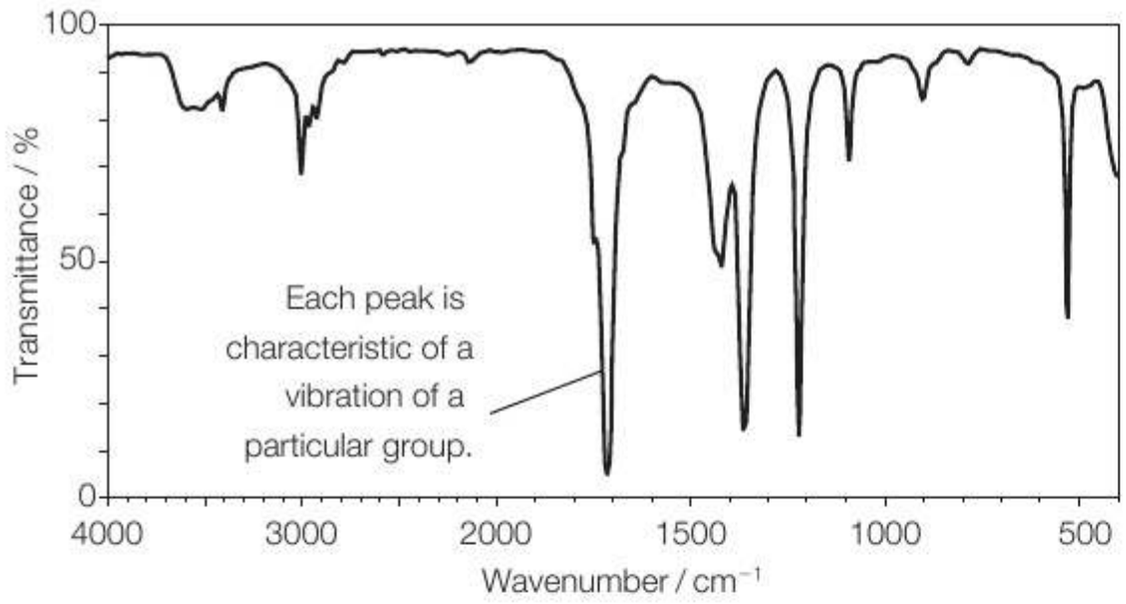
When a compound is irradiated by infrared radiation, the bonds in the molecules absorb radiation from some parts of the spectrum, but not from others.

AXES

The spectrum is normally shown with the vertical axis labelled **transmittance**, shown as a percentage from 0 to 100. A value of 100% transmittance means that 100% of the radiation is transmitted and none is absorbed.

The horizontal axis could be labelled either as frequency or wavelength, but a different unit is used: **wavenumber**. This is the reciprocal of the wavelength, and so it represents frequency. It is usually quoted in the unit cm^{-1}. The numerical scale normally starts at 4000 cm^{-1} and ends at 500 cm^{-1}. It may seem unusual for the numbers to decrease from left to right, but the left to right direction does represent increasing frequency. Another unusual feature is that the scale changes after 2000 cm^{-1}.

Fig B is an example of a typical infrared spectrum.



▲ **fig B** A typical infrared spectrum.

ABSORPTIONS AND THEIR INTENSITIES

You can see that some of the spectrum consists of a wavy but almost horizontal line close to 100%, but at specific wavenumbers there are dips or troughs. These are referred to as absorptions (or sometimes as peaks). The actual transmittance value of an absorption is not very important, but its **intensity** is.

Weak intensities refer to high transmittance values, and strong intensities to low transmittance values.

WAVENUMBER VALUES

The wavenumber values are also very important. The spectrum in **fig B** shows that the absorption with the lowest transmittance occurs at about 1700 cm^{-1}. Sometimes, another important feature to note is whether the absorption is sharp (i.e. a narrow wavenumber range) or broad (a wide wavenumber range).

CHARACTERISTIC ABSORPTIONS

You do not need to remember any of the information in **table A**, as it will be provided in an exam. However, it is important to be familiar with how to use it when interpreting infrared spectra.

WAVENUMBER / cm^{-1}	BOND	FUNCTIONAL GROUP
3750–3200	O–H	alcohol
3500–3300	N–H	amine
3300–2500	O–H	carboxylic acid
3095–3010	C–H	alkene
2962–2853	C–H	alkane
2900–2820 2775–2700	C–H	aldehyde
1740–1720	C=O	aldehyde
1725–1700	C=O	carboxylic acid
1720–1700	C=O	ketone
1669–1645	C=C	alkene

table A Information to help interpret an infrared spectrum.

In the example spectrum in this topic, the strong absorption at 1700 cm^{-1} is caused by C=O, but the compound responsible for this spectrum could be a carboxylic acid or a ketone.

Sometimes a missing absorption is just as useful. In the example, there is no absorption in the 3300–2500 cm^{-1} region, so the compound does not contain an OH group of a carboxylic acid. Therefore, the absorption at 1700 cm^{-1} strongly suggests that the compound is a ketone and not a carboxylic acid.

LEARNING TIP

In an infrared spectrum, remember that the horizontal scale uses wavenumber to represent frequency, and that the scale changes at 2000 cm^{-1}.

CHECKPOINT

1. Explain why hydrogen fluoride absorbs infrared radiation but fluorine does not.
2. A compound has an infrared spectrum that shows a broad absorption centred on 2850 cm^{-1} and a sharp absorption at 1710 cm^{-1}. Suggest what homologous series the compound belongs to.

SUBJECT VOCABULARY

infrared radiation the part of the electromagnetic spectrum with frequencies below that of red light

stretching when a bond absorbs infrared radiation and uses it to alter the length of the bond

transmittance the amount of radiation absorbed at a particular wavenumber

wavenumber the frequency of infrared radiation absorbed by a particular bond in a molecule

intensity the amount of infrared radiation absorbed

10D 4 USING INFRARED SPECTRA

SPECIFICATION REFERENCE
10.22 PART

LEARNING OBJECTIVES

- Be able to use infrared spectra to deduce functional groups present in organic compounds.
- Be able to predict infrared absorptions due to familiar functional groups including wavenumber data.

DIFFERENT WAYS OF USING INFRARED SPECTRA

There are three main ways your understanding of infrared spectra could be tested, and you need to become familiar with all of them. They are:

- predicting the absorptions in the spectrum of an organic compound
- deducing the functional groups in a compound from a list of wavenumbers
- deducing the structure of a compound from wavenumbers and molecular formula.

These all require the use of wavenumber data, like the data shown in **Topic 10D.3**. This will either be provided in the question, or you can use information from the Data booklet provided in the exam. **Table A** is the table from **Topic 10D.3** reproduced here for your convenience.

WAVENUMBER / cm^{-1}	BOND	FUNCTIONAL GROUP
3750–3200	O–H	alcohol
3500–3300	N–H	amine
3300–2500	O–H	carboxylic acid
3095–3010	C–H	alkene
2962–2853	C–H	alkane
2900–2820 2775–2700	C–H	aldehyde
1740–1720	C=O	aldehyde
1725–1700	C=O	carboxylic acid
1720–1700	C=O	ketone
1669–1645	C=C	alkene

table A Information to help interpret an infrared spectrum.

FINGERPRINT REGION

You may come across the term 'fingerprint region'. This term is worth explaining. The table of wavenumbers starts at 3750 and ends at 1645 cm^{-1}, even though there are many absorptions in the infrared region between 1500 to 500 cm^{-1}. Most of the absorptions in this region (1500 and 500 cm^{-1}) result from bending vibrations (not considered in this book) or from absorptions by bonds not listed in the table of wavenumbers.

This region is sometimes referred to as the 'fingerprint region' because, although individual absorptions are not easily recognised, the whole pattern acts like a fingerprint that is slightly different for similar molecules.

PREDICTING THE SPECTRUM OF AN ORGANIC COMPOUND

Suppose you are given the identity of an organic compound. This might be a formula (displayed, skeletal or structural) or a name. You should then be able to predict the wavenumber ranges of the compound's infrared spectrum.

WORKED EXAMPLE 1

What absorptions would you expect to find in the infrared spectrum of propanal?

Once you recognise propanal as an aldehyde, you could predict:

- absorptions in the ranges 2900–2820 and 2775–2700 cm^{-1} resulting from the C–H bond
- absorption in the range 1740–1720 cm^{-1} resulting from the C=O bond.

WORKED EXAMPLE 2

What absorptions would you expect to find in the infrared spectrum of $CH_3CH(OH)CH_3$?

This compound is an alcohol, so you could predict a broad absorption in the range 3750–3200 cm^{-1} resulting from the O–H bond.

DEDUCING THE FUNCTIONAL GROUPS FROM A LIST OF WAVENUMBERS

WORKED EXAMPLE 3

An organic compound has absorptions in the infrared region at these wavenumbers: 3675, 2870 and 1735 cm^{-1}. Which functional groups does it contain?

Answer

O–H (alcohol), C–H and C=O (aldehyde).

WORKED EXAMPLE 4

An organic compound has absorptions in these wavenumber ranges in the infrared region: 3500–3300 and 3300–2500 cm^{-1}. Which functional groups does it contain?

Answer

O–H (carboxylic acid) and N–H (amine): it could be an amino acid.

DEDUCING THE STRUCTURE FROM WAVENUMBERS AND MOLECULAR FORMULA

This is a bit more complicated. Some molecular formulae could represent different combinations of functional groups, so to make the decision you need to consider the actual functional groups from the spectrum and how they could be used in conjunction with the molecular formula.

WORKED EXAMPLE 5

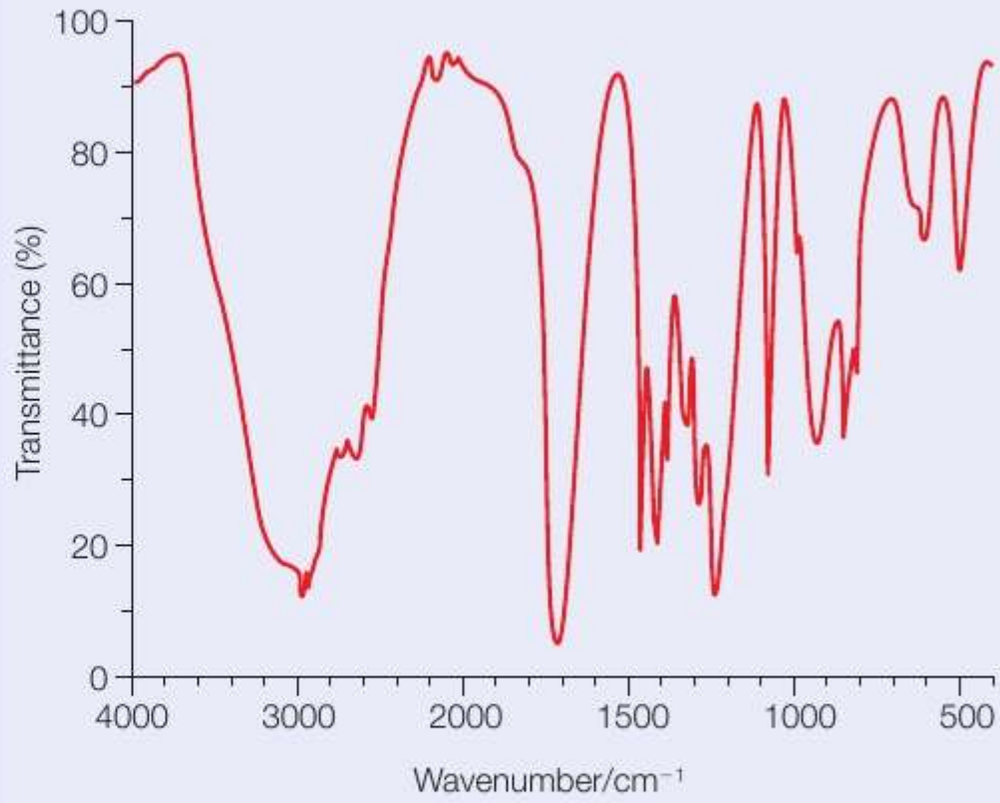
A compound has infrared absorptions at 1730 and 3450 cm^{-1}, and has a molecular formula of $C_2H_4O_2$. Deduce a possible structure for it.

Answer

The functional groups are O–H (alcohol) and C=O (aldehyde). The only structure with the molecular formula $C_2H_4O_2$ that fits is $CH_2(OH)CHO$. Note that it is not CH_3COOH (ethanoic acid) because this would have an absorption in the range 3300–2500 cm^{-1}, not at 3450 cm^{-1}.

WORKED EXAMPLE 6

An organic compound has a molecular formula of $C_3H_6O_2$. Its infrared spectrum is shown below.



▲ **fig A** The infrared spectrum of an unknown compound.

Deduce a possible structure for the compound.

Answer

There is a broad absorption at 3000 cm^{-1} and a narrow absorption at 1700 cm^{-1}. These suggest the presence of O–H (carboxylic acid) and C=O (carboxylic acid). The compound must contain the COOH group, which leaves C_2H_5 to make up the rest of the molecular formula. The only possible structure is CH_3CH_2COOH, which is propanoic acid.

LEARNING TIP

Concentrate on the region between 4000 and 1500 cm^{-1}.
Practise making predictions of the spectra of molecules in **Topics 4**, **5** and **10**.

CHECKPOINT

1. Why is it not easy to use infrared spectra to distinguish between propan-1-ol and propan-2-ol?
2. A compound has the molecular formula C_3H_6O. How can its infrared spectrum be used to show that it has the structure $(CH_3)_2CO$ and not $CH_3C(OH)=CH_2$?